Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/28/2004
PI Name:	Koscheyev, Victor S Ph.D.		
Project Title:	Development of a Finger Calorimeter to Indicate General Thermal Status of Astronauts Onboard and During EVA		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced extravehicular activity		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	55455	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	05/27/2003	**End Date:**	04/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 9 1521		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A specific aim is to design and test a miniaturized sensor and monitoring system (finger calorimeter), applying experimental data assessing the correlation between heat content in the body and heat flux density from the finger area. Year 1 will be devoted to developing and testing different heat flux ring prototypes. In collaboration with industry and the NASA JSC EC/EVA laboratory, in Year 2 a telemetric monitoring system for measuring finger heat flux/temperature will be developed and tested, evaluating several variations of heat flux rings. Year 3 will focus on adjustment of the physiological range of the sensor for compatibility with space equipment, improvement of the sensor and monitoring system, adaptation of the finger calorimeter to the space suit, and ground-based testing. Develop a miniaturized heat flux/temperature finger sensor and monitoring system for eventual use in evaluating thermal status of astronauts during EVA. The heat flux/temperature finger sensor and monitoring system will be a key component in the advanced EVA development of an automatic thermal feedback system to control thermal status of astronauts during EVA.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Applications of the technology for better monitoring of thermal status of personnel in other types of extreme conditions, such as military or HAZMAT situations. Advances in designing support technologies for human protection.
Task Progress:	Year 1 activities focused on: 1. physiological studies to set the parameters in which finger temperature/heat flux will be informative in determining overall body thermal status as monitored by a heat flux/temperature ring on the finger. The eventual goal is to use the heat flux ring as a component of an automatic thermal feedback system to control astronaut thermal status during EVA. Other research included the following: 2. assessment of a heat flux ring prototype under the currently used Phase 6 EVA glove; 3. pilot studies on the efficacy of different physiologically-based glove designs to provide hand comfort under significant chilling of the rest of the body; 4. additional studies on the effectiveness of a head/hand bypass in the tubing of our recently developed shortened liquid cooling/warming garment (SLCWG) in prolonging hand comfort during body chilling; 5. completion of studies of subjects donned in the SLCWG in head down tilt position to assess the possibility of enhancing blood circulation in low extremities to improve comfort. 1. Continued physiological studies and statistical analyses of our physiological data base to assess the physiological boundaries in which finger temperature/heat flux is informative in indicating thermal imbalance of the body, with application for establishing the parameters in which a finger temperature/heat flux ring will be effective. Over the course of Year 1, a total of 37 experiments were conducted with subjects (both male and female) in rest and exercise conditions and in different environmental chamber temperatures. The rest condition studies consisted of changes in inlet water temperature in the SLCWG across stages from 31-33oC; gloves from 23-36oC. Other studies assessed uniform cooling/warming or warming/cooling across stages; and mild cooling (SLCWG inlet water temperature at 20oC); severe cooling (SLCWG inlet water temperature at 8oC). The exercise studies followed the NASA exercise protocol for EVA with subjects in an environmental chamber at 22oC, in either moderate or heavy physical exertion conditions imposed by walking on a treadmill. The results of the analyses of these studies indicated that the most effective range of finger temperature for use with the heat flux ring was between 29.0oC to 34.5oC. 2. Examination of a heat flux ring prototype under the currently used Phase 6 EVA glove. The 4 experiments conducted measured the density of heat flux through the glove thickness compared with the uncovered hands, the placement of the location of the ring, and the possibility to register heat flux if the sensor is placed external to the pressurized rubber bladder. 3. Pilot studies on the efficacy of different physiologically-based glove designs to provide hand comfort under significant chilling of the rest of the body was assessed in a total of 13 experiments, demonstrating that by means of these particular warming gloves, it was possible to extend the time interval before the subject reached a 25oC finger temperature criterion of discomfort. These findings indicate that the physiological design of gloves is a significant advance to conserve heat on the distal part of the extremities and provide comfort. However, variations in the tubing net in the warming gloves have differential efficacy because of their particular contact with the blood vessel network. 4. The effectiveness of a head/hand bypass in the tubing of the SLCWG in maintaining hand temperature was studied by an additional 11 experiments which indicated that it was possible to significantly delay the decline in finger temperature to the 25oC finger temperature criterion of discomfort. These findings demonstrate the potential of combining this biological head/hand heat transfer with electrical heating to minimize energy consumption and prolong finger comfort during EVA. 5. Completion of studies with subjects donned in the SLCWG in an extreme head down tilt position to examine the possibility of enhancing blood circulation to the low extremities as a countermeasure to enhance comfort. The results of this research demonstrated that moderate partial heating of the upper body and thighs improved blood circulation in the feet in simulated microgravity by delivering heat to the lower extremities through the restriction of heat exchange with the environment in the heated body parts. This technique has potential as a supplemental countermeasure onboard for increasing blood circulation to the lower extremities. On the basis of the Year 1 research activities, collaboration with the Mini-Mitter Company, Bend, Oregon, has begun. The company will construct a first prototype heat flux/temperature ring and supporting software that we will test in our laboratory. The ultimate goal is to develop a system for the wireless transmission of data from the heat flux ring to an onboard space station monitor.
Bibliography Type:	Description: (Last Updated: 02/07/2014)
Abstracts for Journals and Proceedings	Koscheyev, V.S.; Leon, G.R.; Treviño, R.C. "Thermal control and physiological design as keys to improve comfort, performance and safety during EVA and onboard." Habitation 9, 92. , Jan-2004
Abstracts for Journals and Proceedings	Leon, G.R.; Koscheyev, V.S.; List, N.; Coca, A. "Comparison of different cooling regimes within a shortened liquid cooling/warming garment on physiological and psychological comfort during exercise." Habitation 9, 91. , Jan-2004
Abstracts for Journals and Proceedings	Koscheyev, V.S.; Leon, G.R.; Coca, A. "Theory and practice of astronaut hand protection against cold in open space." Aviation, Space, and Environmental Medicine 75, Section II, B36. , May-2004
Abstracts for Journals and Proceedings	Coca, A.; Koscheyev, V.S.; Leon, G.R. "Prediction of heat deficit/storage through the dynamic of finger temperature." Annual Meeting of the American College of Sports Medicine, Indianapolis, Indiana, June 4, 2004. Abstract , Jun-2004
Articles in Peer-reviewed Journals	Koscheyev VS, Leon GR, Coca A, List N. "Enhancing circulation to lower limbs during head-down tilt by warming upper body and thighs." Aviation, Space, and Environmental Medicine. in press as of May 2004. , May-2004

Articles in Peer-reviewed Journals	Leon GR, Koscheyev VS, Coca A, List N. "Comparison of different cooling regimes within a shortened liquid cooling/warming garment on physiological and psychological comfort during exercise." Habitation (in press). May 2004. , May-2004
Papers from Meeting Proceedings	Koscheyev, V.S.; Leon, G.R.; Ferl, J.; Coca, A.; Graziosi, D. "Comparison of shortened and standard liquid cooling garments to provide physiological and subjective comfort during EVA." 34th International Conference on Environmental Systems, July 2004. Proceedings of the 34th International Conference on Environmental Systems, July 2004. SAE Technical Paper Series 2004-01-2347. Warrendale, PA : SAE International. , Jul-2004